|INCH-POUND|

MIL-R-83726/18E(USAF)
9 August 1991
SUPERSEDING
MIL-R-83726/18D(USAF)
19 February 1988

MILITARY SPECIFICATION SHEET

RELAYS, HYBRID, TIME DELAY (ON RELEASE), TRACK MOUNTED, TYPE IIA, CLASS B, 10 AMPERES, 2PDT, HERMETICALLY SEALED, FIXED TIME, 0.1 TO 300 SECONDS

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-83726.

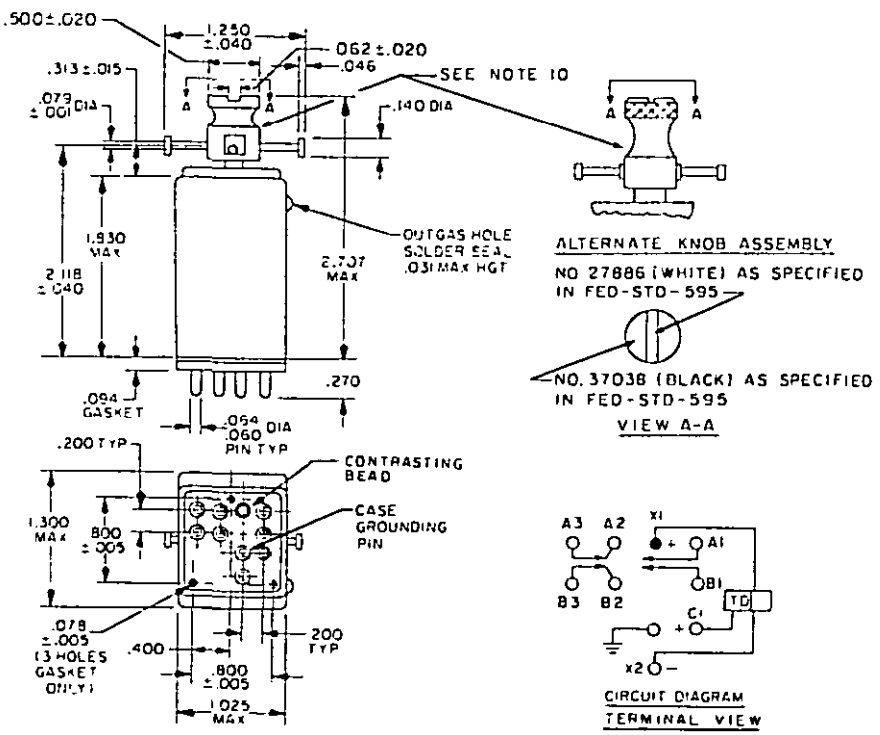


FIGURE 1. Outline dimensions and configuration of relay.

(E) denotes changes

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Inches	mm
.001	0.03
.005	0.13
.015	0.25
.031	0.79
.040	1.02
.046	1.17
.060	1.52
.064	1.63
.078	1.98
.079	2.01
.094	2.39
.140	3.56
.200	5.08
.270	6.86
.313	7.95
.400	10.16
.800	20.32
1.025	26.04
1.250	31.75
2.118	53.80
2.707	68.76

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
3. Metric equivalents are given for general information only.
4. Terminal numbers shall not appear on the relay header and there shall be a legible circuit diagram on relay which identifies each terminal location specified.
5. For relay mounting track see MIL-R-6106/3 and for relay bracket and socket assembly see MIL-R-6106/5.
6. Gasket material: Methylvinyl silicone (red). Shore hardness A40.
7. Terminal plating - Gold in accordance with MIL-G-45204 type II, class 1. Underplating: nickel, 50 to 150 microinches thick.
8. Enclosure finish color: FED-STD-595, green 34449.
9. Coil suppression: Back EMF 42 V dc maximum.
10. Knob and post shall be nickel plated (100 microinches thick minimum) 303 stainless steel.

FIGURE 1. Outline dimensions and configuration of relay - Continued.

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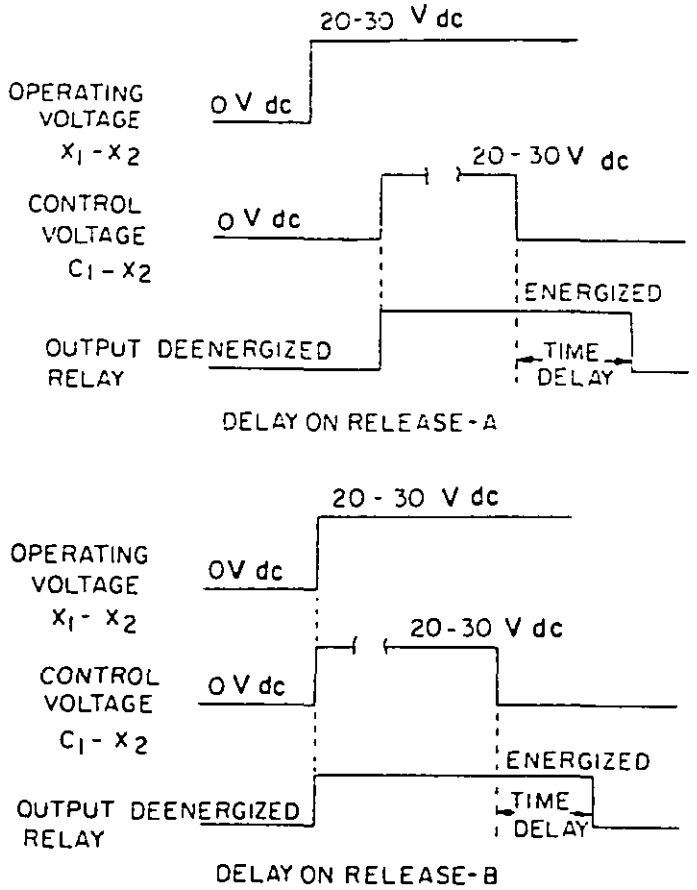


FIGURE 2. Timing diagrams (both A and B conditions are applicable).

TABLE I. Dash numbers and time delay characteristics.

Dash number 1/ 2/	Time delay (seconds) ±10% 3/
-1000	0.1
-1001	1.0
-5001	5.0
-1602	16.0
-6002	60.0
-1803	180.0
-3003	300.0

1/ Additional time delay relays within the 0.1- to 300-second delay range are available. To establish Part or Identifying Numbers (PIN's) not listed in table I (see "PIN" herein).

2/ PIN's having a suffix letter A, B, or C following the dash number impose more stringent EMI requirements than PIN's with a suffix letter W, X, or Y. For Government logistical support the suffixed part numbers will be supplied (see "PIN" herein).

3/ Ten percent of nominal value under all combination of input voltage and environmental extremes.

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REQUIREMENTS:

Contact data:

Configuration: 2PDT.

Life/load ratings: See table II.

TABLE II. Life load ratings (relay case grounded).

Type of load	Life (cycles)	Amperes	
		28 V dc	115/200 1 and 3 phase 400 Hz
Resistive	100,000	10	10
Inductive	50,000	8	8
Motor	100,000	4	4
Lamp	100,000	2	2
Mechanical life (reduced current)	400,000	2.5	2.5

Contact voltage drop:

Initial: 0.150 volt.

After life tests: 0.175 volt.

Minimum current: Applicable in accordance with MIL-R-6106.

Contact bounce: 1 millisecond (maximum).

Overload:

DC: 40 amperes.

AC: 60 amperes.

Rupture: 1/

DC: 50 amperes.

AC: 80 amperes.

Input data:

Duty rating: Continuous.

Maximum voltage (over temperature range): 30 V dc.

Nominal voltage (over temperature range): 28 V dc.

Minimum voltage (over temperature range): 20 V dc.

Minimum voltage high-temperature test: 20 V dc.

Minimum voltage continuous current tests: 20 V dc.

Maximum coil current at 25°C: 0.150 ampere.

1/ Not applicable if time current relay characteristics at 25°C are required, or if the MS27401 relay being used is qualified to MIL-R-6106.

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Operate time: 10 milliseconds (maximum).

Recycle time: 50 milliseconds maximum (over the total temperature range and with both minimum and maximum input voltages). Relay will recycle if control voltage is removed from and reapplied to the control terminal C1 during the timing cycles.

Electrical data:

Insulation resistance at 500 V dc: 2/

Initial: 100 megohms (minimum).

After life or environmental tests: 100 megohms (minimum).

Dielectric strength (sea level): 2/

	Initial	After life tests
Coil to case 3/	1,050 V rms	1,050 V rms
All other points. 4/	1,250 V rms	1,250 V rms

Dielectric strength (altitude) (80,000 feet): 2/ 5/

	Initial	After life tests
Coil to case. 3/	350 V rms	350 V rms
All other points. 4/	350 V rms	350 V rms

Transients (at terminals X1, X2, and C1):

Transient voltage limits (input): In accordance with MIL-STD-461 and MIL-STD-462 (methods CS01, CS02, CS06, RS02, and RS03), for PIN's with a suffix letter W, X, or Y and PIN's with a suffix letter A, B, or C, the same limits apply except the field intensity limits for RS03 are as specified in table III. Susceptibility testing shall not cause the relay to change state nor affect the timing interval beyond allowable tolerance limits.

TABLE III. RS03 levels for suffix letter A, B, or C PIN's.

Frequency	RS03 level (volt/meter)
14 kHz to 200 MHz	20
200 MHz to 450 MHz	65
450 MHz to 1 GHz	20
1 GHz to 12 GHz	100

Transient voltage limits (output): In accordance with MIL-STD-461 and MIL-STD-462 (methods CE03 and RE02), voltage spikes generated by time delay relays shall not exceed ±42 volts during the following time periods starting at the instant of opening of the relay coil circuit:

2/ Terminals X1, X2 and C1 connected together during this test.
3/ Leakage current limited to 50 microamperes.
4/ Leakage current limited to 25 microamperes.
5/ Relay not installed in its bracket and socket assembly (M6106/5-004).

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PIN's with a suffix letter W, X, or Y: 50 microseconds.

PIN's with a suffix letter A, B, or C: 500 microseconds.

Monitoring of spike voltages shall be by a device (such as a memory type oscilloscope) that can detect, measure, display, and record 0.25 to 1000 volts over the applicable time durations specified herein. Switches used to control the time delay relays during voltage-spike measurements shall not be EMI suppressed.

Time current relay characteristics at 25°C: 6/ Relays shall sustain five applications (make and carry only) of power concurrently on adjacent poles at each of five different current levels for the time durations shown in table IV. Separate relays shall be tested at 28 V dc and 115/200 V ac, 400 Hz, 3 phase. Cooling time between successive applications shall be 30 minutes. The test shall be performed on both normally open and normally closed contacts of each relay. There shall be no failures or evidence of welding or sticking, and relays shall pass contact voltage drop at conclusion.

TABLE IV. Time-current relay characteristics at 25°C.

1	15 A - 1 hour
2	50 A - 5.0 seconds
3	100 A - 1.2 seconds
4	250 A - 0.2 second
5	350 A - 0.1 second

CAUTION: Compare with time-current characteristics of associated circuit protective device.

Environmental data (relay installed in bracket and socket assembly (M6106/5-004) and secured in mounting track (M6106/3)):

Temperature range (operating): -70°C to +125°C.

Maximum altitude rating: 80,000 feet.

Shock G level: 50 g's.

Duration: 10 milliseconds.

Maximum duration contact opening: 10 microseconds.

Maximum duration contact closing: 1 microsecond.

Vibration (sinusoidal):

G level: 30 g's.

Frequency range: 10 Hz to 3,000 Hz.

6/ Not applicable if the MS27401 relay being used is qualified to the time current relay characteristics at 25°C requirements of MIL-R-6106.

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Ⓔ Vibration (random): Applicable in accordance with MIL-STD-202, method 214, test condition 18.

Power spectral density: 0.4 g^2/Hz.

Frequency range: 50 Hz to 2,000 Hz.

Ⓔ Duration: 15 minutes each plane.

Acceleration: 15 g's.

Seal: Applicable to nonhermetically-sealed relays and hermetically-sealed (potted) relays.

Physical data:

Dimensions and configuration: *See figure 1.*

Weight: 0.34 pound (maximum).

Knob strength: The rod to knob junction shall be capable of resisting an axial pull of 35 pounds without degradation.

Ⓔ Marking: See MIL-R-83726. In addition, relays shall be marked with the ESDS identifier as specified in MIL-STD-1285.

Ⓔ ESDS protection program: An ESDS protection program shall be implemented within 6 months of the date of revision E to this document. The manufacturer shall establish and maintain an ESD control program in accordance with MIL-STD-1686 for mission critical equipment. Evidence of such compliance shall be verified by the qualifying activity of this specification as a prerequisite for qualification and continued qualification. This program shall be documented by an ESD control plan which must be under document control. As a minimum, this plan must address the identification of ESDS sub-components and end items, facilities, training, design protection, handling procedures, marking, cleaning, preservation, packaging, and quality assurance. A model ESD control program is available from the qualifying activity and may be used as a guideline. Further guidance for ESD control is available from the EOS/ESD Association and the Electronics Industry Association (EIA). This requirement is applicable to all manufacturers who handle ESDS component parts and materials in the relay manufacturing or testing process. This requirement is not limited to manufacturers qualifying ESDS end items.

Ⓔ ESDS verification: As a part of qualification or qualification after redesign, ESD testing shall be done in accordance with method 3015 of MIL-STD-883 modified to test at 16,000 volts. Testing at lower voltage levels is not required. This testing shall be accomplished as part of the group III for qualification inspection and as part of the group C inspection.

Ⓔ ESDS preservation and packaging: Relays shall be preserved and packaged in such a manner as to ensure that the integrity of ESD sensitive relays is not diminished. ESD sensitive relays shall be preserved and packaged in accordance with the requirements of MIL-STD-1686.

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PIN: Consists of the prefix M83726/18-, a four-digit dash number (expressed in milliseconds), and a suffix letter from table V (indicating the extent of EMI requirements and quality level) as follows:

	M 83726/18- 1001 A
Military designation	M
Specification sheet number	83726/18-
The first three digits are significant; the fourth digit is the number of zeros to follow the first three digits. The time is expressed in milliseconds and converted to seconds. (See examples following table V.)	1001
Suffix letter from table V. 7/	A

TABLE V. Suffix letter for EMI requirements and quality levels.

Quality level	Suffix letters for relays with less stringent EMI requirements	Suffix letters for relays with more stringent EMI requirements
W	W	A
Y	Y	B
X	X	C

Examples:

M83726/18-1000W - 100 millisecond time delay, W quality level.
M83726/18-1001A - 1 second time delay with more stringent EMI requirements and W quality level.
M83726/18-3003B - 300 second time delay with more stringent EMI requirements and Y quality level.

7/ Any relays numbered prior to the date of this specification without a quality level indication shall be considered interchangeable (store and issue) with the W quality level.

CONCLUDING MATERIAL

Custodian:
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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